

Using ROS 2 Jazzy and Gazebo on Ubuntu 24.04 — What We Tried and What Happened

Overview

ROS 2 Jazzy is very new (released May 2025). The Gazebo packages that work with ROS 2 (`gazebo_ros_pkgs`) aren't officially ready for Jazzy yet, so it's tricky to set up Gazebo with ROS 2 Jazzy on Ubuntu 24.04.

1. Trying to Download Gazebo Packages for Jazzy

I tried to get the Gazebo packages by cloning the Jazzy branch from GitHub:

```
git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git
-b jazzy
```

```
fatal: Remote branch jazzy not found in upstream origin
```

I also tried the `rolling` branch:

```
git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git
-b rolling
```

And got a similar error:

```
fatal: Remote branch rolling not found in upstream origin
```

Why? Because neither Jazzy nor Rolling branches exist in the Gazebo repo yet. So, Jazzy support isn't officially there.

2. What Worked: Using the Humble Branch Instead

Since Jazzy is mostly compatible with the Humble version, I cloned the Humble branch like this:

```
cd ~/ros2_gz_ws/src
git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git
-b humble
```

This worked! I could build the Gazebo packages from source and use them with ROS 2 Jazzy.

3. Problems With Dependencies on Ubuntu 24.04

ROS 2 Foxy (an older ROS 2 version) was built for Ubuntu 20.04 and needs some specific libraries like `libpython3.8`, OpenCV 4.2, and others.

But these libraries aren't available on Ubuntu 24.04, so trying to install Foxy or Humble packages directly on Ubuntu 24.04 fails.

Example error:

```
vb net
```

```
E: Unable to locate package ros-foxy-desktop
```

4. Why Foxy and Humble Don't Work Well on Ubuntu 24.04

Ubuntu 24.04 is newer than the versions Foxy and Humble were made for.

Many of the libraries these ROS versions depend on are missing or incompatible with Ubuntu 24.04.

You can try to build Foxy or Humble from source, but it's complicated and often doesn't work smoothly.

5. What About ROS 2 Iron?

ROS 2 Iron is made for Ubuntu 24.04, so it should be a perfect fit.

But as of now (mid-2025), Iron packages aren't available to install via apt yet.

Trying to install Iron fails with this error:

```
sudo apt install ros-iron-desktop
```

```
vb net
```

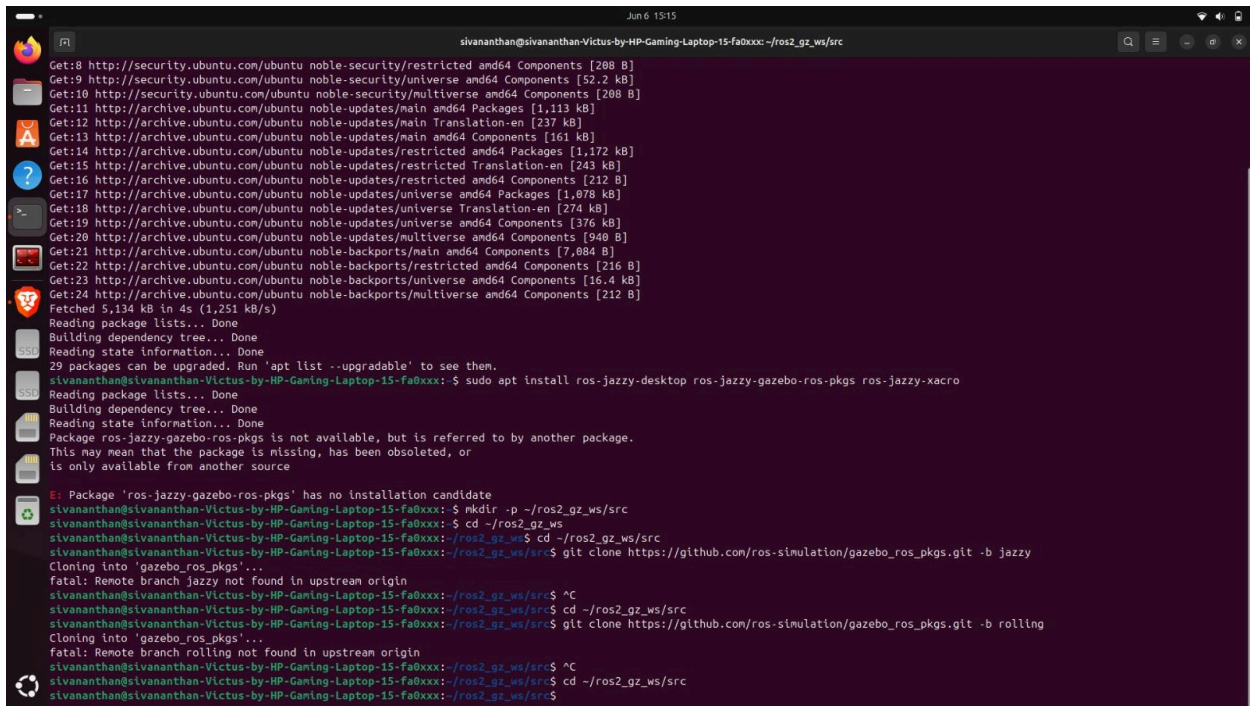
```
E: Unable to locate package ros-iron-desktop
```

6. Best Solution for Now: Use ROS 2 Humble in Docker (But We Tried and It Failed)

The easiest way to avoid the compatibility problems is to run ROS 2 Humble inside a Docker container based on Ubuntu 22.04.

This keeps your Ubuntu 24.04 system clean and gives you a working ROS and Gazebo setup.

However, We tried this approach as well, but it didn't work for us.

A terminal window screenshot showing the process of installing ROS 2 on Ubuntu 24.04. The terminal output shows the system checking for updates and attempting to install 'ros-jazzy-gazebo-ros-pkgs'. It fails with an error: 'Package 'ros-jazzy-gazebo-ros-pkgs' has no installation candidate'. The user then attempts to clone the repository from GitHub, but fails with the error: 'fatal: Remote branch jazzy not found in upstream origin'. The terminal window title is 'sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~/ros2_gz_ws/src'.

```
Jun 6 15:15
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~/ros2_gz_ws/src

Get:8 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [288 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [288 B]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,113 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [237 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,172 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [243 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,078 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [274 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [376 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,884 B]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.4 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 5,134 kB in 4s (1,251 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
29 packages can be upgraded. Run 'apt list --upgradable' to see them.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt install ros-jazzy-desktop ros-jazzy-gazebo-ros-pkgs ros-jazzy-xacro
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package ros-jazzy-gazebo-ros-pkgs is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'ros-jazzy-gazebo-ros-pkgs' has no installation candidate
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ mkdir -p ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ cd ~/ros2_gz_ws
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git -b jazzy
Cloning into 'gazebo_ros_pkgs'...
fatal: Remote branch jazzy not found in upstream origin
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ ^C
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git -b rolling
Cloning into 'gazebo_ros_pkgs'...
fatal: Remote branch rolling not found in upstream origin
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ ^C
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$
```

```
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~/ros2_gz_ws/src
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,113 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [237 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,172 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [243 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,078 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [274 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [376 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,884 B]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.4 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 5,134 kB in 4s (1,251 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
29 packages can be upgraded. Run 'apt list --upgradable' to see them.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt install ros-jazzy-desktop ros-jazzy-gazebo-ros-pkgs ros-jazzy-xacro
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package ros-jazzy-gazebo-ros-pkgs is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
E: Package 'ros-jazzy-gazebo-ros-pkgs' has no installation candidate
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ mkdir -p ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ cd ~/ros2_gz_ws
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git -b jazzy
Cloning into 'gazebo_ros_pkgs'...
fatal: Remote branch jazzy not found in upstream origin
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ ^C
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git -b rolling
Cloning into 'gazebo_ros_pkgs'...
fatal: Remote branch rolling not found in upstream origin
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ ^C
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ cd ~/ros2_gz_ws/src
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$ git clone https://github.com/ros-simulation/gazebo_ros_pkgs.git -b humble
Cloning into 'gazebo_ros_pkgs'...
fatal: Remote branch humble not found in upstream origin
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/ros2_gz_ws/src$
```

URDF to Gazebo Integration

GitHub - ros-simulation/

ROS2 Jazzy assignment - C

(12) WhatsApp

github.com/ros-simulation/gazebo_ros_pkgs

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<> Code Issues 295 Pull requests 96 Actions Projects Wiki Security Insights

noetic-devel 63 Branches 101 Tags

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Code

J-rivero 2.9.3 cd9cc95 · 4 months ago 1,010 Commits

gazebo_dev	2.9.3	4 months ago
gazebo_msgs	2.9.3	4 months ago
gazebo_plugins	2.9.3	4 months ago
gazebo_ros	2.9.3	4 months ago
gazebo_ros_control	2.9.3	4 months ago
gazebo_ros_pkgs	2.9.3	4 months ago
.gitignore	updating .gitignore	12 years ago
CONTRIBUTING.md	Removed all trailing whitespace	9 years ago
README.md	Add deprecation warning to README (#1567)	4 months ago
SENSORS.md	Include an entry in SENSORS.md	7 years ago

README

gazebo_ros_pkgs

Warning

Deprecation

About

Wrappers, tools and additional API's for using ROS with Gazebo

wiki.ros.org/gazebo_ros_pkgs

Readme

Activity

Custom properties

823 stars

44 watching

784 forks

Report repository

Releases

101 tags

Packages

No packages published

Contributors 83



+ 69 contributors

Languages


```
Jun 6 15:33
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
Depends: libopencv-videoio4.2 (>= 4.2.0+dfsg) but it is not installable
ros-foxy-libstatistics-collector : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-lifecycle-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-logging-demo : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-nav-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-pcl-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-pendulum-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rcl-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rcl-logging-splog : Depends: libspdlog1 (>= 1.5.0) but it is not installable
ros-foxy-rclpy : Depends: libpython3.8 (>= 3.8.2) but it is not installable
Depends: ros-foxy-unique-identifier-msgs but it is not going to be installed
ros-foxy-rosbag2-converter-default-plugins : Depends: ros-foxy-rmw-fastrtps-cpp but it is not installable
ros-foxy-rosbag2-cpp : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rosbag2-storage : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rosbag2-transport : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rosgraph-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rtt-image-view : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: ros-foxy-cv-bridge but it is not going to be installed
Depends: ros-foxy-image-transport but it is not going to be installed
Depends: ros-foxy-qt-gui-cpp but it is not going to be installed
Depends: ros-foxy-rtt-gui-cpp but it is not going to be installed
ros-foxy-rviz-common : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rviz-default-plugins : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
Depends: ros-foxy-map-msgs but it is not going to be installed
ros-foxy-sensor-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-shape-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-statistics-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-std-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-std-srvs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-stereo-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-tf2-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-tf2-py : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-trajectory-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-turtlesim : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-visualization-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
E: Unable to correct problems, you have held broken packages.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~$ sudo apt update
Hit:1 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://packages.ros.org/ros2/ubuntu focal InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... 8%
```

```
Jun 6 15:35
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
Depends: libopencv-imgcodecs4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-videoio4.2 (>= 4.2.0+dfsg) but it is not installable
ros-foxy-intra-process-demo : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-highgui4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-videoio4.2 (>= 4.2.0+dfsg) but it is not installable
ros-foxy-libstatistics-collector : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-lifecycle-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-logging-demo : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-nav-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-pcl-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-pendulum-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rcl-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rcl-logging-splog : Depends: libspdlog1 (>= 1.5.0) but it is not installable
ros-foxy-rclpy : Depends: libpython3.8 (>= 3.8.2) but it is not installable
Depends: ros-foxy-unique-identifier-msgs but it is not going to be installed
ros-foxy-rosbag2-converter-default-plugins : Depends: ros-foxy-rmw-fastrtps-cpp but it is not installable
ros-foxy-rosbag2-cpp : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rosbag2-storage : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rosbag2-transport : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rosgraph-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-rtt-image-view : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
Depends: ros-foxy-cv-bridge but it is not going to be installed
Depends: ros-foxy-image-transport but it is not going to be installed
Depends: ros-foxy-qt-gui-cpp but it is not going to be installed
Depends: ros-foxy-rtt-gui-cpp but it is not going to be installed
ros-foxy-rviz-common : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
ros-foxy-rviz-default-plugins : Depends: libtinyxml2-6a (>= 5.0.0) but it is not installable
Depends: ros-foxy-map-msgs but it is not going to be installed
ros-foxy-sensor-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-shape-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-statistics-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-std-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-std-srvs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-stereo-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-tf2-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-tf2-py : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-trajectory-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-turtlesim : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-visualization-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
E: Unable to correct problems, you have held broken packages.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~$ ^C
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~$
```

```
libbpl-dev libavformat-dev libavcodec-dev libswresample-dev liburiparser-dev
libavfilter9
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following upgrades have been deferred due to phasing:
cloud-init ibverbs-providers libibverbs-dev libibverbs1 libqt5qml5 libqt5qmlmodels5 libqt5qmlworkerscript5 libqt5quick5 libqt5quickparticles5 libqt5quickshapes5 libqt5quicktest5
libqt5quickwidgets5 librdmacm164 libtraceevent1 libtraceevent1-plugin openssl-client qml-module-qt-labs-folderlistmodel qml-module-qt-labs-settings qml-module-qtqml
qml-module-qtqml-models2 qml-module-qtquick-layouts qml-module-qtquick-window2 qml-module-qtquick2 qt5-qmltooling-plugins qtdeclarative5-dev qtdeclarative5-dev-tools
ubuntu-drivers-common update-notifier update-notifier-common
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo apt install ros-foxy-desktop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages could not be installed. This may mean that you have
requested an impossible situation or if you are using the unstable
distribution that some required packages have not yet been created
or been moved out of Incoming.
The following information may help to resolve the situation:

The following packages have unmet dependencies:
ros-foxy-action-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
                        Depends: ros-foxy-unique-identifier-msgs but it is not going to be installed
ros-foxy-action-tutorials-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-actionlib-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-builtin-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-composition-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-demo-nodes-cpp-native : Depends: ros-foxy-fastrtsp but it is not installable
                                Depends: ros-foxy-rmw-fastrtsp-cpp but it is not installable
ros-foxy-depthimage-to-laserscan : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
                                Depends: ros-foxy-image-geometry but it is not going to be installed
ros-foxy-diagnostic-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-example-interfaces : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-geometry-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-image-tools : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
                        Depends: libopencv-highgui4.2 (>= 4.2.0+dfsg) but it is not installable
                        Depends: libopencv-imgcodecs4.2 (>= 4.2.0+dfsg) but it is not installable
                        Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
                        Depends: libopencv-video4.2 (>= 4.2.0+dfsg) but it is not installable
ros-foxy-intra-process-demo : Depends: libopencv-core4.2 (>= 4.2.0+dfsg) but it is not installable
                                Depends: libopencv-highgui4.2 (>= 4.2.0+dfsg) but it is not installable
                                Depends: libopencv-imgproc4.2 (>= 4.2.0+dfsg) but it is not installable
                                Depends: libopencv-video4.2 (>= 4.2.0+dfsg) but it is not installable
ros-foxy-libstatistics-collector : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-lifecycle-msgs : Depends: libpython3.8 (>= 3.8.2) but it is not installable
ros-foxy-logging-demo : Depends: libpython3.8 (>= 3.8.2) but it is not installable
```

```
Jun 6 15:40
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo apt update
[sudo] password for sivananthan:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 http://packages.ros.org/ros2/ubuntu focal InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
29 packages can be upgraded. Run 'apt list --upgradable' to see them.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo apt install locales
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
locales is already the newest version (2.39-0ubuntu8.4).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo locale-gen en_US.UTF-8
Generating locales (this might take a while)...
en_US.UTF-8... done
Generation complete.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ export LANG=en_US.UTF-8
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.49.2).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-Fa0xxx:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-C to cancel.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://packages.ros.org/ros2/ubuntu focal InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... 0%
```



```
Jun 6 15:41
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-C to cancel.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://packages.ros.org/ros2/ubuntu focal InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt update && sudo apt install curl gnupg lsb-release
Hit:1 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:2 http://packages.ros.org/ros2/ubuntu focal InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
29 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
gnupg is already the newest version (2.4.4-2ubuntu17.2).
lsb-release is already the newest version (12.0-2).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring
.gpg
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros
.org/ros2/ubuntu $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/ros2.list
deb [arch=amd64 signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu noble main
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt update
Hit:1 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://packages.ros.org/ros2/ubuntu noble InRelease [4,676 B]
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:7 http://packages.ros.org/ros2/ubuntu noble/main amd64 Packages [1,585 kB]
88% [7 Packages 1,198 kB/1,585 kB 76%]
179 kB/s 2.1
```

```
Jun 6 15:46
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx: ~
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt update
[sudo] password for sivananthan:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://packages.osrfoundation.org/gazebo/ubuntu-stable noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://packages.ros.org/ros2/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
29 packages can be upgraded. Run 'apt list --upgradable' to see them.
sivananthan@sivananthan-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~$ sudo apt install -y build-essential cmake git python3-colcon-common-extensions python3-pip python3-rosdep python3-rosinst
-generator python3-vcstool wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
build-essential set to manually installed.
cmake is already the newest version (3.28.3-1build7).
cmake set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.2).
git set to manually installed.
python3-colcon-common-extensions is already the newest version (0.3.0-100).
python3-colcon-common-extensions set to manually installed.
python3-rosdep is already the newest version (0.25.1-1).
python3-rosdep set to manually installed.
python3-vcstool is already the newest version (0.3.0-1).
python3-vcstool set to manually installed.
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following additional packages will be installed:
  python3-wheel
The following NEW packages will be installed:
  python3-pip python3-rosinstall-generator python3-wheel
0 upgraded, 3 newly installed, 0 to remove and 29 not upgraded.
Need to get 1,382 kB of archives.
After this operation, 7,310 kB of additional disk space will be used.
Get:1 http://packages.ros.org/ros2/ubuntu noble/main amd64 python3-rosinstall-generator all 0.1.23-1 [11.0 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-wheel all 0.42.0-2 [53.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip all 24.0+dfsg-1ubuntu1.1 [1,317 kB]
Fetched 1,382 kB in 3s (444 kB/s)
Selecting previously unselected package python3-wheel.
(Reading database ... 20235 files and directories currently installed.)
```



```
Jun 6 15:59
root@06bb2ceaf83b: ~/ros2_ws

sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x root@06bb2ceaf83b: ~/ros2_ws x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x

Skip end-of-life distro "galactic"
Skip end-of-life distro "groovy"
Add distro "humble"
Skip end-of-life distro "hydro"
Skip end-of-life distro "indigo"
Skip end-of-life distro "iron"
Skip end-of-life distro "jade"
Add distro "jazzy"
Add distro "kilted"
Skip end-of-life distro "kinetic"
Skip end-of-life distro "lunar"
Skip end-of-life distro "melodic"
Skip end-of-life distro "noetic"
Add distro "rolling"
updated cache in /root/.ros/rosdep/sources.cache
root@06bb2ceaf83b:~/ros2_ws# rosdep install --from-paths src --ignore-src -r -y
#All required rosdeps installed successfully
root@06bb2ceaf83b:~/ros2_ws# colcon build

Summary: 0 packages finished [0.23s]
root@06bb2ceaf83b:~/ros2_ws# source install/setup.bash
root@06bb2ceaf83b:~/ros2_ws# gazebo --verbose
bash: gazebo: command not found
root@06bb2ceaf83b:~/ros2_ws# gz --verbose
bash: gz: command not found
root@06bb2ceaf83b:~/ros2_ws# apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:3 http://packages.ros.org/ros2/ubuntu jammy InRelease [4682 B]
Get:4 http://packages.ros.org/ros2/ubuntu jammy/main Sources [1698 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [4476 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2984 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [47.7 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1246 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1553 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3295 kB]
74k [11 Packages store 0 B] [16 Packages 6433 B/3295 kB 0%] [4 Sources 861 kB/1698 kB 51%] 3738 kB/s 2s
```

```
Jun 6 16:02
root@06bb2ceaf83b: ~/ros2_ws

sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x root@06bb2ceaf83b: ~/ros2_ws x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x

Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1553 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3295 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [55.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [4630 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.2 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [83.2 kB]
Get:21 http://packages.ros.org/ros2/ubuntu jammy/main amd64 Packages [1672 kB]
Fetched 42.1 MB in 14s (2965 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@06bb2ceaf83b:~/ros2_ws# apt install -y \
> gazebo11 \
> libgazebo11-dev \
> ros-humble-gazebo-ros-pkgs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package gazebo11
E: Unable to locate package libgazebo11-dev
root@06bb2ceaf83b:~/ros2_ws# sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://packages.ros.org/ros2/ubuntu jammy InRelease
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@06bb2ceaf83b:~/ros2_ws# sudo apt install -y \
> ros-humble-gazebo-ros-pkgs \
> ros-humble-ros2-control-demos \
> ros-humble-ros2-control \
> ros-humble-ros2-controllers \
> ros-humble-ign-gazebo
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package ros-humble-ros2-control-demos
E: Unable to locate package ros-humble-ign-gazebo
root@06bb2ceaf83b:~/ros2_ws#
```

```
Jun 6 16:04
root@06bb2ceaf33b: ~/ros2_ws

sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x root@06bb2ceaf33b: ~/ros2_ws x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x

conerr-dev fonts-liberation freeglut3 freeglut3-dev gazebo gazebo-common gazebo-plugin-base krb5-multidev libass9 libavc1394-0 libavdevice-dev libavdevice58 libavfilter-dev
libavfilter7 libbz2b0 libbsd-dev libcaca0 libccd-dev libccd2 libcdio-cdda2 libcdio-paranoia2 libcdio19 libdart-collision-bullet6.12 libdart-dev libdart-external-convhull-3d-dev
libdart-external-odelcpsolver-dev libdart-external-odelcpsolver6.12 libdart-util16.12 libdart6.12 libfcl-dev libfcl0.7 libflite1 libfreeimage-dev libfreeimage3 libgazebo-dev
libgazebo11 libgssrpc4 libgts-dev libiec61883-0 libignition-common3 libignition-common3-av-dev libignition-common3-av3 libignition-common3-core-dev libignition-common3-dev
libignition-common3-events3 libignition-common3-graphics-dev libignition-common3-graphics3 libignition-common3-profiler3 libignition-fuel-tools4.4 libignition-fuel-tools4-dev
libignition-msgs5-5 libignition-msgs5-dev libignition-transport8-8 libignition-transport8-dev libjack-jackd2-0 libjxr0 libkadm5clnt-mt12 libkadm5srv-mt12 libkdb5-10 libkrb5-dev
libllv-0-0 libndev libmysofa1 libnorm-dev liboctomap-dev liboctomap1.9 libode8 libogre-1.9-dev libogre-1.9.0v5 libopenal-data libopenal1 libpgm-dev libpocketsphinx3
libpostproc-dev libpostproc55 libprotobuf-dev libprotobuf-lite23 libprotoc-dev libprotoc23 libqt-propertybrowser5 libqt5-qt5-6 libqt5-qt5-dev libraw20 librubberband2 libsampleate0
libstdformat9-9 libstdformat9-dev libstd-0-0 libstdbody-dev libstdbody3.6 libstring2 libstdiun-dev libstd-0-0 libspinxbase3 libspnav0 libstrat0-0 libtar-dev libtar0
libtinyobjloader-dev libtinyobjloader1 liburdfdom-model3.0 liburdfdom-world3.0 liburdfdom-stab1.1 libzimg2 libzfp-dev libzip4 libzmq3-dev libzzip-0-13 pocketsphinx-en-us
protobuf-compiler ros-humble-camera-calibration-parsers ros-humble-camera-info-manager ros-humble-gazebo-dev ros-humble-gazebo-msgs ros-humble-gazebo-plugins sdfmat9-sdf

Suggested packages:
doc-base gazebo-doc krb5-doc krb5-user libgts-doc ignition-tools jackd2 libnorm-doc ogre-1.9-doc libogre-1.9.0v5-dbg libportaudio2 serdi sordi spacenavd protobuf-mode-el

The following NEW packages will be installed:
conerr-dev fonts-liberation freeglut3 freeglut3-dev gazebo gazebo-common gazebo-plugin-base krb5-multidev libass9 libavc1394-0 libavdevice-dev libavdevice58 libavfilter-dev
libavfilter7 libbz2b0 libbsd-dev libcaca0 libccd-dev libccd2 libcdio-cdda2 libcdio-paranoia2 libcdio19 libdart-collision-bullet6.12 libdart-dev libdart-external-convhull-3d-dev
libdart-external-odelcpsolver-dev libdart-external-odelcpsolver6.12 libdart-util16.12 libdart6.12 libfcl-dev libfcl0.7 libflite1 libfreeimage-dev libfreeimage3 libgazebo-dev
libgazebo11 libgssrpc4 libgts-dev libiec61883-0 libignition-common3 libignition-common3-av-dev libignition-common3-av3 libignition-common3-core-dev libignition-common3-dev
libignition-common3-events3 libignition-common3-graphics-dev libignition-common3-graphics3 libignition-common3-profiler3 libignition-fuel-tools4.4 libignition-fuel-tools4-dev
libignition-msgs5-5 libignition-msgs5-dev libignition-transport8-8 libignition-transport8-dev libjack-jackd2-0 libjxr0 libkadm5clnt-mt12 libkadm5srv-mt12 libkdb5-10 libkrb5-dev
libllv-0-0 libndev libmysofa1 libnorm-dev liboctomap-dev liboctomap1.9 libode8 libogre-1.9-dev libogre-1.9.0v5 libopenal-data libopenal1 libpgm-dev libpocketsphinx3
libpostproc-dev libpostproc55 libprotobuf-dev libprotobuf-lite23 libprotoc-dev libprotoc23 libqt-propertybrowser5 libqt5-qt5-6 libqt5-qt5-dev libraw20 librubberband2 libsampleate0
libstdformat9-9 libstdformat9-dev libstd-0-0 libstdbody-dev libstdbody3.6 libstring2 libstdiun-dev libstd-0-0 libspinxbase3 libspnav0 libstrat0-0 libtar-dev libtar0
libtinyobjloader-dev libtinyobjloader1 liburdfdom-model3.0 liburdfdom-world3.0 liburdfdom-stab1.1 libzimg2 libzfp-dev libzip4 libzmq3-dev libzzip-0-13 pocketsphinx-en-us
ros-humble-gazebo-ros-pkgs sdfmat9-sdf
0 upgraded, 118 newly installed, 0 to remove and 0 not upgraded.
Need to get 138 MB of archives.
After this operation, 378 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 libslang2 amd64 2.3.2-5build4 [468 kB]
Get:2 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-camera-calibration-parsers amd64 3.1.11-1jammy.20250429.201740 [54.5 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-liberation all 1:1.07.4-11 [822 kB]
Get:4 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-camera-info-manager amd64 3.1.11-1jammy.20250429.202052 [42.3 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/universe amd64 freeglut3 amd64 2.8.1-6 [74.0 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/universe amd64 freeglut3-dev amd64 2.8.1-6 [126 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/main amd64 libavc1394-0 amd64 0.5.4-Sbuild2 [17.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libass9 amd64 1:0.15.2-1 [97.5 kB]
Get:9 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-gazebo-dev amd64 3.9.0-1jammy.20250128.005836 [8050 B]
Get:10 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-gazebo-msgs amd64 3.9.0-1jammy.20250325.190319 [492 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libbz2b0 amd64 3.1.0+dfsg-2.2build1 [10.2 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libflite1 amd64 2.2-3 [13.7 MB]
8% [12 libflite1 8460 kB/13.7 MB 62%] [Waiting for headers]
```

```
Jun 6 16:04
root@06bb2ceaf33b: ~/ros2_ws

sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x root@06bb2ceaf33b: ~/ros2_ws x sivananthan@sivananthan-Victus... x sivananthan@sivananthan-Victus... x

Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libqt5-qt5-6 amd64 6.1.4-2 [418 kB]
Get:55 http://archive.ubuntu.com/ubuntu jammy/universe amd64 sdfmat9-sdf all 9.7.0+ds-1 [57.1 kB]
Get:56 http://archive.ubuntu.com/ubuntu jammy/universe amd64 liburdfdom-model3.0 amd64 3.0.1-1 [43.9 kB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libstdformat9-9 amd64 9.7.0+ds-1 [442 kB]
Get:58 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libstdbody3.6 amd64 3.6.1+dfsg-7build1 [3324 kB]
Get:59 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libspnav0 amd64 0.2.3-1 [8552 B]
Get:60 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libtar0 amd64 1.2.20-8ubuntu0.22.04.1 [19.7 kB]
Get:61 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libtinyobjloader1 amd64 2.0.0-rc5+dfsg1-Sbuild1 [63.9 kB]
Get:62 http://archive.ubuntu.com/ubuntu jammy/universe amd64 liburdfdom-world3.0 amd64 3.0.1-1 [42.9 kB]
Get:63 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libgazebo11 amd64 11.10.2+dfsg-1 [8734 kB]
Get:64 http://archive.ubuntu.com/ubuntu jammy/universe amd64 gazebo-common all 11.10.2+dfsg-1 [35.7 MB]
Get:65 http://archive.ubuntu.com/ubuntu jammy/universe amd64 gazebo amd64 11.10.2+dfsg-1 [4095 kB]
Get:66 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libdart-util16.12 amd64 6.12.1+dfsg4-11build2 [285 kB]
Get:67 http://archive.ubuntu.com/ubuntu jammy/universe amd64 gazebo-plugin-base amd64 11.10.2+dfsg-1 [2165 kB]
Get:68 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgssrpc4 amd64 1.19.2-2ubuntu0.7 [58.4 kB]
Get:69 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkdb5-10 amd64 1.19.2-2ubuntu0.7 [40.5 kB]
Get:70 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkadm5srv-mt12 amd64 1.19.2-2ubuntu0.7 [54.7 kB]
Get:71 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkadm5clnt-mt12 amd64 1.19.2-2ubuntu0.7 [41.8 kB]
Get:72 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 conerr-dev amd64 2.1-1.46.5-2ubuntu1.2 [41.0 kB]
Get:73 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 krb5-multidev amd64 1.19.2-2ubuntu0.7 [125 kB]
Get:74 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libpostproc-dev amd64 7:4.4.2-8ubuntu0.22.04.1 [60.9 kB]
Get:75 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libavfilter-dev amd64 7:4.4.2-8ubuntu0.22.04.1 [1732 kB]
Get:76 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libavdevice-dev amd64 7:4.4.2-8ubuntu0.22.04.1 [97.3 kB]
Get:77 http://archive.ubuntu.com/ubuntu jammy/main amd64 libndev amd64 1.0.4-1build1 [40.0 kB]
Get:78 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbsd-dev amd64 0.11.5-1 [168 kB]
Get:79 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libccd-dev amd64 2.1-2 [10.5 kB]
Get:80 http://archive.ubuntu.com/ubuntu jammy/universe amd64 liboctomap-dev amd64 1.9.7+dfsg-3 [1164 kB]
Get:81 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfcl-dev amd64 0.7.0-3 [235 kB]
Get:82 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libdart-external-odelcpsolver-dev amd64 6.12.1+dfsg4-11build2 [15.2 kB]
Get:83 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libdart-external-convhull-3d-dev amd64 6.12.1+dfsg4-11build2 [13.8 kB]
Get:84 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libdart-dev amd64 6.12.1+dfsg4-11build2 [201 kB]
Get:85 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libqt5-qt5-dev amd64 6.1.4-2 [98.7 kB]
Get:86 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotoc23 amd64 3.12.4-1ubuntu7.22.04.2 [662 kB]
Get:87 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotobuf-lite23 amd64 3.12.4-1ubuntu7.22.04.2 [209 kB]
Get:88 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-gazebo-plugins amd64 3.9.0-1jammy.20250429.225502 [1621 kB]
Get:89 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotobuf-dev amd64 3.12.4-1ubuntu7.22.04.2 [1347 kB]
Get:90 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libprotoc-dev amd64 3.12.4-1ubuntu7.22.04.2 [999 kB]
Get:91 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libfreeimage-dev amd64 3.18.0+ds2-6ubuntu5.1 [371 kB]
Get:92 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libogre-1.9-dev amd64 1.9.0+dfsg1-12.1ubuntu1 [955 kB]
Get:93 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libtar-dev amd64 1.2.20-8ubuntu0.22.04.1 [37.8 kB]
Get:94 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpgm-dev amd64 5.3.128+dfsg-2 [193 kB]
Get:95 http://archive.ubuntu.com/ubuntu jammy/main amd64 libstdiun-dev amd64 1.0.18-1build2 [185 kB]
74% [95 libstdiun-dev 16.6 kB/185 kB 9%] [88 ros-humble-gazebo-plugins 68.4 kB/1621 kB 4%] 8285 kB/s 4...
```



```
Jun 6 16:04
root@06bb2ceaf83b: ~/ros2_ws

Preparing to unpack .../081-fonts-liberation_1k3a1.07.4-11_all.deb ...
Unpacking fonts-liberation (1:1.07.4-11) ...
Selecting previously unselected package freeglut3:amd64.
Preparing to unpack .../082-freeglut3_2.8.1-6_amd64.deb ...
Unpacking freeglut3:amd64 (2.8.1-6) ...
Selecting previously unselected package freeglut3-dev:amd64.
Preparing to unpack .../083-freeglut3-dev_2.8.1-6_amd64.deb ...
Unpacking freeglut3-dev:amd64 (2.8.1-6) ...
Selecting previously unselected package libavc1394-0:amd64.
Preparing to unpack .../084-libavc1394-0_0.5.4-5build2_amd64.deb ...
Unpacking libavc1394-0:amd64 (0.5.4-5build2) ...
Selecting previously unselected package libass9:amd64.
Preparing to unpack .../085-libass9_1k3a0.15.2-1_amd64.deb ...
Unpacking libass9:amd64 (1:0.15.2-1) ...
Selecting previously unselected package libbs2b0:amd64.
Preparing to unpack .../086-libbs2b0_3.1.0+dfsg-2build1_amd64.deb ...
Unpacking libbs2b0:amd64 (3.1.0+dfsg-2build1) ...
Selecting previously unselected package libflite1:amd64.
Preparing to unpack .../087-libflite1_2.2-3_amd64.deb ...
Unpacking libflite1:amd64 (2.2-3) ...
Selecting previously unselected package libserd-0-0:amd64.
Preparing to unpack .../088-libserd-0-0_0.30.10-2_amd64.deb ...
Unpacking libserd-0-0:amd64 (0.30.10-2) ...
Selecting previously unselected package libsord-0-0:amd64.
Preparing to unpack .../089-libsord-0-0_0.16.0-2_amd64.deb ...
Unpacking libsord-0-0:amd64 (0.16.0-2) ...
Selecting previously unselected package libstrat0-0:amd64.
Preparing to unpack .../090-libstrat0-0_0.6.8-1_amd64.deb ...
Unpacking libstrat0-0:amd64 (0.6.8-1) ...
Selecting previously unselected package libllv-0-0:amd64.
Preparing to unpack .../091-libllv-0-0_0.24.12-2_amd64.deb ...
Unpacking libllv-0-0:amd64 (0.24.12-2) ...
Selecting previously unselected package libnysofa1:amd64.
Preparing to unpack .../092-libnysofa1_1.2.1+dfsg-1_amd64.deb ...
Unpacking libnysofa1:amd64 (1.2.1+dfsg-1) ...
Selecting previously unselected package libspnbase3:amd64.
Preparing to unpack .../093-libspnbase3_0.8.5prealpha+1-13build1_amd64.deb ...
Unpacking libspnbase3:amd64 (0.8.5prealpha+1-13build1) ...
Selecting previously unselected package libpocketsphinx3:amd64.
Preparing to unpack .../094-libpocketsphinx3_0.8.0+realSprealpha+1-14ubuntu1_amd64.deb ...
Unpacking libpocketsphinx3:amd64 (0.8.0+realSprealpha+1-14ubuntu1) ...
Progress: [ 6%] [#####.....]
```

```
Jun 6 16:06
root@06bb2ceaf83b: ~/ros2_ws

Setting up libdart6.12:amd64 (6.12.1+dfsg4-11build2) ...
Setting up libgnition-common3-graphics3:amd64 (3.14.0+dfsg-5) ...
Setting up libdart-collision-bullet6.12:amd64 (6.12.1+dfsg4-11build2) ...
Setting up libllv-0-0:amd64 (0.24.12-2) ...
Setting up libfreeimage-dev (3.18.0+ds2-6ubuntu5.1) ...
Setting up librs-multidev:amd64 (1.19.2-2ubuntu0.7) ...
Setting up libgnition-common3-graphics-dev:amd64 (3.14.0+dfsg-5) ...
Setting up libogre-1.9-dev (1.9.0+dfsg1-12.1ubuntu1) ...
Setting up libsdformat9-dev:amd64 (9.7.0+ds-1) ...
Setting up libgnition-msgs5-dev:amd64 (5.8.1+ds-2) ...
Setting up libkrb5-dev:amd64 (1.19.2-2ubuntu0.7) ...
Setting up libdart-util6.12:amd64 (6.12.1+dfsg4-11build2) ...
Setting up libavfilter7:amd64 (7:4.4.2-0ubuntu0.22.04.1) ...
Setting up libzmq3-dev:amd64 (4.3.4-2) ...
Setting up libignition-fuel-tools4-dev:amd64 (4.4.0+ds-3build1) ...
Setting up libavfilter-dev:amd64 (7:4.4.2-0ubuntu0.22.04.1) ...
Setting up libignition-transport8-dev:amd64 (8.2.1+ds-2) ...
Setting up libavdevice58:amd64 (7:4.4.2-0ubuntu0.22.04.1) ...
Setting up libavdevice-dev:amd64 (7:4.4.2-0ubuntu0.22.04.1) ...
Setting up libgazebo11:amd64 (11.10.2+dfsg-1) ...
Setting up gazebo (11.10.2+dfsg-1) ...
Setting up libignition-common3-av-dev:amd64 (3.14.0+dfsg-5) ...
Setting up libignition-common3-dev:amd64 (3.14.0+dfsg-5) ...
Setting up gazebo-plugin-base (11.10.2+dfsg-1) ...
Setting up libdart-external-combustion-3d-dev:amd64 (6.12.1+dfsg4-11build2) ...
Setting up libdart-dev:amd64 (6.12.1+dfsg4-11build2) ...
Setting up libgazebo-dev (11.10.2+dfsg-1) ...
Setting up ros-humble-gazebo-dev (3.9.0-1jammy.20250128.005836) ...
Setting up ros-humble-gazebo-ros (3.9.0-1jammy.20250429.212904) ...
Setting up ros-humble-gazebo-plugins (3.9.0-1jammy.20250429.225502) ...
Setting up ros-humble-gazebo-ros-pkgs (3.9.0-1jammy.20250429.235927) ...
Processing triggers for libc-bin (2.35-0ubuntu3.10) ...
Processing triggers for mailcap (3.78+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
root@06bb2ceaf83b:~/ros2_ws# sudo apt install ros-humble-ros-ign-msgs ros-humble-ros-ign-common ros-humble-ros-ign-transport ros-humble-ros-ign-gazebo
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package ros-humble-ros-ign-msgs
E: Unable to locate package ros-humble-ros-ign-common
E: Unable to locate package ros-humble-ros-ign-transport
root@06bb2ceaf83b:~/ros2_ws#
```

```
Jun 6 16:09
root@06bb2ceaf83b:~/ros2_ws

sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x root@06bb2ceaf83b:~/ros2_ws x sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x

2JqGsvQym3sAPnWVA68
=xH9H
-----END PGP PUBLIC KEY BLOCK-----
E: The list of sources could not be read.
root@06bb2ceaf83b:~/ros2_ws# cat /etc/apt/sources.list.d/ros2.list
deb [signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu jammy main
root@06bb2ceaf83b:~/ros2_ws# sudo rm /etc/apt/sources.list.d/ros2.list
root@06bb2ceaf83b:~/ros2_ws# echo "deb [signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu jammy main" | sudo tee /etc/apt/sources.list.d/ros2.list
deb [signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu jammy main
root@06bb2ceaf83b:~/ros2_ws# sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg
root@06bb2ceaf83b:~/ros2_ws# sudo apt update
E: Conflicting values set for option Signed-By regarding source http://packages.ros.org/ros2/ubuntu/ jammy: /usr/share/keyrings/ros-archive-keyring.gpg != -----BEGIN PGP PUBLIC KEY BLOCK-----
nQINBFzvjPvBEADy8l1Vv071VH5gUeSyzT6mMvUmlV3KArBaJz9KCRmgPKh7Jc
VfrQhE8L/HV7LOfoLI9H2GWYH8qN5ER8lcaBxvG3ZvX7l9nAZP0T2Xx3GT3tro
uSoCR+SyH9XpNpUduQUSvJ2eqNyb9B/Hph30ntjG30jSNa9K0F5b8TK1h0TCPh4
K/AY0jzT60pHFxU6ytlF5I47ZKsnTUhtpCsKucQ1CKLyirndZ3V3k70YaoqZ5SrG
aIoAWLx2H8j7sAHmqS29N9jV9mo135d+dTdLBI0PxtLHzE9IPaX+ctdSURPnp+
TwR99LxglpIG6hLuvDMAaxiqFBB/JF3XJ80BakF56nHrMH2WqQxRb1ITl0rkQoz
pwNEF2Bv0+Jvs1UFEDVz5a8xexQhst/RnKrThLct3LOcVBNoQAQRbvVvBHPj0/p
V5cYeUljZ5wPhyFkaEVLCLaVWqa6PisylmqyjsruPCMLURLsQoQABCL8bwxK7UT
hM6GtH6tGLVZ8SRI2RlFIn2oudzV5l+8oRgFr9yVcwyoFT6JCl0qkwlDw5ZP1pk/
/5nuexC6LYqduHUS5NokzpzF56qHFTYSP5tz4KHJf5jD1Ktly3mKYFvF5PVV
okGdcUvz2he1f3jxb8N1B/jvtk0bImfcs08n7H+4E92N1sX/LSb0KQARAOAB
tCZPCGVuIF3jYvN90aWnzIDxpbmZvQ6ZcncZv0K5KX8pb24ub3JnPkCVAQTAQGA
PgIbAWILCQhAgYVcgIKcWlEfGIDAQIeAQIXgYbHMHb3JmUt6IaLFtYpQ1vur
F8ZUB0JoEhoCBQkUHzWAAoJEPQ1vurF8ZUv1AP/ZgID+uw7pw3MdpEvny3pliz
JedX4Y+ut+5c2NcFkpUc3LGS0v9ly4ZpNQTKIn9yB6dXgar77EKpA1GVmLU75JA
LyftVtjeyQcre2F7Z08u2LXw8Red52AsHkh/dtctgLSGQ1JdTd0d0n06cszZFVa
sC1fDRkLizGv8KE8UfDKYMGlxOgnvQZrb9OLqRsoj10aDzN0X3NjK1LxIS3+udY
po0k2Bn9GyrNngIrYlNqyPBHYKwGHbQ3xvAK92LJ2l/n6X4U8r6sMdDE7QDw4j
FMDrxCBXNCL4cFPKKR1qadtJy9f1CtpkyqLUuUSC6AULSE0V+7Y30SpKn8mpIK5
VNBv12JRI2tI0eAmw2qyB0WANCIF/OqkUFCLUpJKyR30XxwJxotIP+HmdaBw
HcnHFIu2th/eqgK656tzcVx3JxYfneF92MN7FEfQsKZhl/jt31pE5HrpdAaz
opRInRw7yEtz3VaaehmUUR2gevPQMzBRaA4IqMLDUv5j3jYf8c1VUtblTYc
/a1BLM/MoInly3aUfDahzhTr6zz+ur6BFRnFW56M3NOVLrenN31BNX2kTKh0Z
QJSSCKL3uDuQnPaZT2BZUwpwfe7QYRvVqHf0YB2N1LavyNwlyfInCQlAh+Q9eme
2JqGsvQym3sAPnWVA68
=xH9H
-----END PGP PUBLIC KEY BLOCK-----
E: The list of sources could not be read.
root@06bb2ceaf83b:~/ros2_ws#
```

```
Jun 6 16:11
root@06bb2ceaf83b:~/ros2_ws

sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x root@06bb2ceaf83b:~/ros2_ws x sivananthan@sivananthan-Victus-... x sivananthan@sivananthan-Victus-... x

/a1BLM/MoInly3aUfDahzhTr6zz+ur6BFRnFW56M3NOVLrenN31BNX2kTKh0Z
QJSSCKL3uDuQnPaZT2BZUwpwfe7QYRvVqHf0YB2N1LavyNwlyfInCQlAh+Q9eme
2JqGsvQym3sAPnWVA68
=xH9H
-----END PGP PUBLIC KEY BLOCK-----
E: The list of sources could not be read.
root@06bb2ceaf83b:~/ros2_ws# sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key | gpg --dearmor | sudo tee /usr/share/keyrings/ros-archive-keyring.gpg > /dev
/null
root@06bb2ceaf83b:~/ros2_ws# sudo rm /usr/share/keyrings/ros-archive-keyring.gpg
root@06bb2ceaf83b:~/ros2_ws# sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key | gpg --dearmor | sudo tee /usr/share/keyrings/ros-archive-keyring.gpg > /dev
/null
root@06bb2ceaf83b:~/ros2_ws# sudo apt update
E: Conflicting values set for option Signed-By regarding source http://packages.ros.org/ros2/ubuntu/ jammy: /usr/share/keyrings/ros-archive-keyring.gpg != -----BEGIN PGP PUBLIC KEY BLOCK-----
nQINBFzvjPvBEADy8l1Vv071VH5gUeSyzT6mMvUmlV3KArBaJz9KCRmgPKh7Jc
VfrQhE8L/HV7LOfoLI9H2GWYH8qN5ER8lcaBxvG3ZvX7l9nAZP0T2Xx3GT3tro
uSoCR+SyH9XpNpUduQUSvJ2eqNyb9B/Hph30ntjG30jSNa9K0F5b8TK1h0TCPh4
K/AY0jzT60pHFxU6ytlF5I47ZKsnTUhtpCsKucQ1CKLyirndZ3V3k70YaoqZ5SrG
aIoAWLx2H8j7sAHmqS29N9jV9mo135d+dTdLBI0PxtLHzE9IPaX+ctdSURPnp+
TwR99LxglpIG6hLuvDMAaxiqFBB/JF3XJ80BakF56nHrMH2WqQxRb1ITl0rkQoz
pwNEF2Bv0+Jvs1UFEDVz5a8xexQhst/RnKrThLct3LOcVBNoQAQRbvVvBHPj0/p
V5cYeUljZ5wPhyFkaEVLCLaVWqa6PisylmqyjsruPCMLURLsQoQABCL8bwxK7UT
hM6GtH6tGLVZ8SRI2RlFIn2oudzV5l+8oRgFr9yVcwyoFT6JCl0qkwlDw5ZP1pk/
/5nuexC6LYqduHUS5NokzpzF56qHFTYSP5tz4KHJf5jD1Ktly3mKYFvF5PVV
okGdcUvz2he1f3jxb8N1B/jvtk0bImfcs08n7H+4E92N1sX/LSb0KQARAOAB
tCZPCGVuIF3jYvN90aWnzIDxpbmZvQ6ZcncZv0K5KX8pb24ub3JnPkCVAQTAQGA
PgIbAWILCQhAgYVcgIKcWlEfGIDAQIeAQIXgYbHMHb3JmUt6IaLFtYpQ1vur
F8ZUB0JoEhoCBQkUHzWAAoJEPQ1vurF8ZUv1AP/ZgID+uw7pw3MdpEvny3pliz
JedX4Y+ut+5c2NcFkpUc3LGS0v9ly4ZpNQTKIn9yB6dXgar77EKpA1GVmLU75JA
LyftVtjeyQcre2F7Z08u2LXw8Red52AsHkh/dtctgLSGQ1JdTd0d0n06cszZFVa
sC1fDRkLizGv8KE8UfDKYMGlxOgnvQZrb9OLqRsoj10aDzN0X3NjK1LxIS3+udY
po0k2Bn9GyrNngIrYlNqyPBHYKwGHbQ3xvAK92LJ2l/n6X4U8r6sMdDE7QDw4j
FMDrxCBXNCL4cFPKKR1qadtJy9f1CtpkyqLUuUSC6AULSE0V+7Y30SpKn8mpIK5
VNBv12JRI2tI0eAmw2qyB0WANCIF/OqkUFCLUpJKyR30XxwJxotIP+HmdaBw
HcnHFIu2th/eqgK656tzcVx3JxYfneF92MN7FEfQsKZhl/jt31pE5HrpdAaz
opRInRw7yEtz3VaaehmUUR2gevPQMzBRaA4IqMLDUv5j3jYf8c1VUtblTYc
/a1BLM/MoInly3aUfDahzhTr6zz+ur6BFRnFW56M3NOVLrenN31BNX2kTKh0Z
QJSSCKL3uDuQnPaZT2BZUwpwfe7QYRvVqHf0YB2N1LavyNwlyfInCQlAh+Q9eme
2JqGsvQym3sAPnWVA68
=xH9H
-----END PGP PUBLIC KEY BLOCK-----
E: The list of sources could not be read.
root@06bb2ceaf83b:~/ros2_ws#
```